

**Manuscript Title: “Polymer-Mediated Pathway Control of CO₂ Laser-Treated TiO₂
in Ambient Air for Enhanced Solar Vapor Generation”**

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Sincerely,

Seung-Mo Lee, Prof. Dr.

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	Name	Affiliation	E-mail
1			
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Author information

	Name	Address/Phone/Email
1	Dao Thi Dung	Korea Institute of Machinery & Materials (KIMM), 156 Gajungbukno, Yuseong-gu, Daejeon 305-343, South Korea Email: daothidung94mt@gmail.com .
2	Do Van Lam	Korea Institute of Machinery & Materials (KIMM), 156 Gajungbukno, Yuseong-gu, Daejeon 305-343, South Korea Email: daothidung94mt@gmail.com .
3	Prof. Dr. Seung-Mo Lee	Korea University of Science and Technology (UST), 217 Gajeong-ro, Yuseong-gu, Daejeon 305-333, Korea Email: sm.lee@kimm.re.kr

Polymer-Mediated Pathway Control of CO₂ Laser-Treated TiO₂ in Ambient Air for Enhanced Solar Vapor Generation

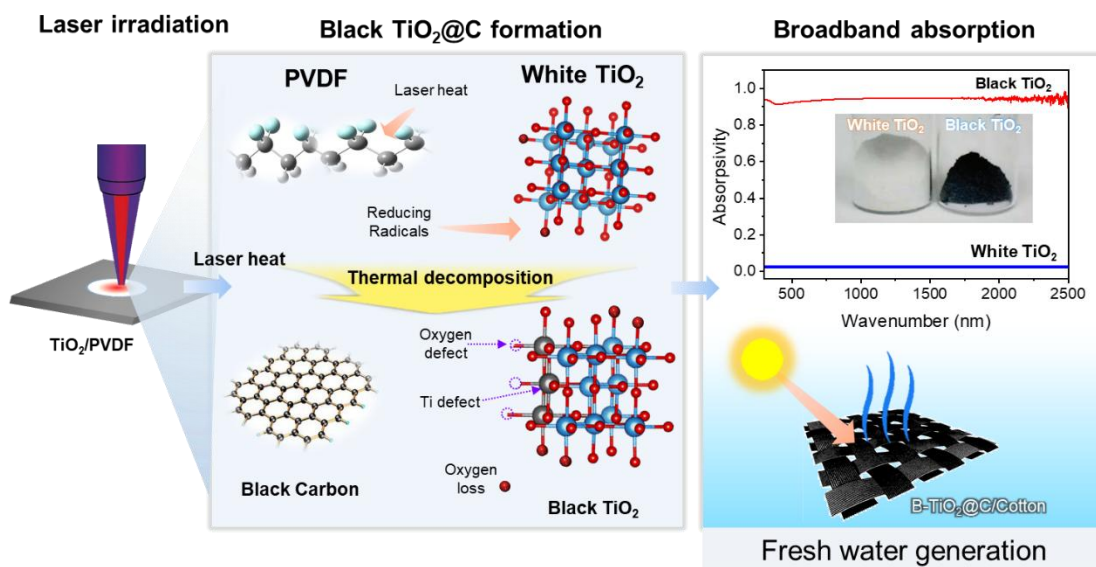
Dao Thi Dung^{a,b}, Do Van Lam^{a,b}, and Seung-Mo Lee^{a,b,*}

^a Korea Institute of Machinery and Materials (KIMM), 156 Gajeongbuk-ro, Yuseong-gu, Daejeon 34103, South Korea.

^b University of Science and Technology (UST), 217 Gajeong-ro, Yuseong-gu, Daejeon 34113, South Korea.

* Corresponding author: Seung-Mo Lee (sm.lee@kimm.re.kr).

TOC Graphic and Summary



PVDF-assisted CO₂ laser irradiation rapidly converts white TiO₂ into defect-rich black TiO₂@C hybrids, enabling broadband solar absorption for fresh water generation.

ABSTRACT

Black TiO_2 has emerged as a promising material for solar energy conversion due to its defect-induced visible-light absorption and antibacterial properties. However, scalable and rapid synthesis remains challenging. Conventional approaches, including hydrogenation, chemical reduction, and solvothermal treatments, require high temperatures, additional chemical reagents, and long reaction times. Herein, we introduce a poly(vinylidene fluoride)–assisted CO_2 laser strategy for the rapid fabrication of defect-rich black $\text{TiO}_2@\text{C}$ hybrid materials. Poly(vinylidene fluoride) serves as both a laser absorber and a chemically reducing precursor. Poly(vinylidene fluoride) strongly absorbs $10.6\ \mu\text{m}$ laser radiation to generate localized photothermal heat, which provides energy to break Ti-O bonding. Besides, the polymer decomposes into reactive species, including H, CF_2 , and carbonaceous fragments, which promote Ti^{4+} reduction and oxygen-defect formation. As a result, white TiO_2 on the PVDF matrix can be transformed into black TiO_2 embedded on black carbon hybrids within seconds using CO_2 laser irradiation. The obtained product exhibits broadband solar absorption with a solar absorptance of 94.1% and rapid photothermal heating to 68°C under 1-sun irradiation. In solar-driven desalination, a cotton-based flat photothermal coated with $\text{TiO}_2@\text{C}$ achieves a high evaporation rate of $1.51\ \text{kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$ with additional antibacterial activity. This work provides a rapid, solvent-free, and scalable route for defect formation of TiO_2 for solar vapor generation applications.

1. Introduction

Titanium dioxide (TiO_2) nanomaterials have long been considered one of the most promising metal oxide semiconductors for solar energy conversion due to their chemical stability, abundance, low cost, and environmental friendliness [1–3]. However, all three crystal phases of TiO_2 (anatase, rutile, and brookite) have a wide electronic band gap of approximately 3.0–3.2 eV [4,5], limiting light absorption primarily to the ultraviolet region (100–400 nm). This intrinsic limitation affects the performance of the TiO_2 -based photocatalytic, photothermal absorber system [6]. Over the past decade, numerous efforts have been made to modify the optical properties of TiO_2 through non-metal and metal doping [7–9]. Among these strategies, black TiO_2 -rich oxygen defect, a modified form of TiO_2 , has attracted particular attention [10,11]. These defect-induced changes in the electronic structure enable improvements in solar energy harvesting performance [12,13].

Black TiO_2 has gained significant attention, and numerous studies have explored its synthesis through various methods, including hydrogenation, chemical reduction, electrochemical reduction, solvothermal reduction, and plasma treatment [14,15]. These current technologies can effectively produce black TiO_2 with a narrow band gap and enhanced absorption across the solar spectrum [16,17]. They often suffer significant limitations, including a long reaction time, high-temperature conditions, additional chemical requirements, and high overall cost (Table S1). Laser irradiation has been proposed as a promising alternative due to its easy operation, rapid synthesis, and high product purity. However, a fundamental challenge arises from the fact that white TiO_2 exhibits strong reflectance in the visible light (0.3–0.7 μm) and near-infrared (0.7–2.5 μm), leading to poor CO₂ laser absorption. Due to this limitation, recent studies convert white TiO_2 to black TiO_2 using laser irradiation in a wet environment and still need a long processing time - roughly 2 hours [18–20]. Therefore, developing efficient methods to transform white TiO_2 to black TiO_2 directly remains an important and unresolved problem.

To overcome this limitation, polyvinylidene fluoride (PVDF) was introduced as an infrared-absorbing supported material. The C-F, C-C, and C-H bonds in PVDF molecules exhibit strong absorption near 10.6 μm , enabling rapid photothermal heating when mixed with white TiO_2 . Under intense laser irradiation, PVDF decomposes, generating highly reactive species, including H, CF_2 , and carbonaceous fragments [21]. These species act as powerful reducing agents that promote the transformation of Ti^{4+} to Ti^{3+} , accelerate the removal of lattice oxygen, and create oxygen vacancy defects—key structural features of black TiO_2 . Furthermore, carbon residues from PVDF decomposition may deposit on the TiO_2 surface, thereby contributing additional defect states that enhance visible-light absorption. By supplying both localized heat and reducing chemical species, PVDF significantly improves the efficiency of CO_2 -laser-induced conversion, enabling a fast and direct transformation of white TiO_2 into black TiO_2 .

In this study, we demonstrate a rapid and direct CO_2 Laser method for synthesizing black TiO_2 from white TiO_2 /PVDF hybrid under ambient conditions. Applying laser irradiation to TiO_2 /PVDF with the PVDF doping of 10%, the black TiO_2 was successfully synthesized. The resulting laser-induced defects, including surface disorder and Ti^{3+} species, significantly enhance visible light absorption. The obtained product exhibits broadband solar absorption, with a solar absorptance of 94%, and rapidly reaches 68°C under 1-sun irradiation. In solar-driven desalination, a cotton-based flat photothermal coated with $\text{TiO}_2\text{@C}$ achieves a high evaporation rate of 1.51 $\text{kg}\cdot\text{m}^{-2}\cdot\text{h}^{-1}$ with additional antibacterial activity. These laser methods present a promising avenue for advanced energy applications, like photocatalysts and light harvesting.

2. Results and discussion

2.1. *Sample preparation*

In this work, a commercial CO₂ laser operating at 10.64 μm was used as a rapid, localized energy source to induce the transformation of white TiO₂ into a rich defect TiO₂. A TiO₂/PVDF hybrid thin film was first prepared and uniformly pasted onto an aluminum foil substrate with a thickness of 0.5 mm, as schematically illustrated in [Figure 1A](#). The aluminum foil plays as a mechanical support the laser processing.

Upon laser irradiation in ambient air, the TiO₂/PVDF hybrid film changes within the irradiated region within few seconds ([Figure S1](#)). The intense, localized laser energy induced thermal decomposition and carbonization of the PVDF matrix, yielding a conductive black carbon phase. Concurrently, the laser irradiation provided sufficient energy to partially break Ti–O bonds within the TiO₂ lattice. This process led to the reduction of Ti⁴⁺ to Ti³⁺ species and the generation of oxygen vacancies, which are well-known defect states in TiO₂. These structural and electronic modifications collectively led to a dramatic change in the material's optical appearance, with the irradiated area transitioning from white to black, as shown in [Video supporting S1](#).

The pronounced color change strongly suggests a substantial alteration in the electronic structure of TiO₂, specifically an increase in its solar absorption. White TiO₂ is a wide-band-gap semiconductor and therefore reflects a large portion of the solar spectrum ([Figure 1B](#)). [Figure 1B](#) demonstrates that the laser-converted TiO₂@C exhibits markedly enhanced broadband optical absorptivity across the Vis–NIR region compared to pristine TiO₂, approaching that of black carbon, which is indicative of improved light-harvesting capability. The resulting black TiO₂@C coating can be applied to a floating substrate to serve as an efficient photothermal layer, enabling effective conversion of solar energy into heat for interfacial water evaporation ([Figure 1C](#)). It offers strong potential for solar-driven desalination and freshwater production applications.

2.2. Polymer screening

To determine the most effective polymer additive for CO₂ laser-assisted conversion of TiO₂ coatings, a comparative screening study was conducted using several representative polymers, including poly(vinylidene fluoride) (PVDF), polyethylene (PE), poly(methyl methacrylate) (PMMA), polydimethylsiloxane (PDMS), and poly(vinyl alcohol) (PVA), together with a no-polymer control. [Figure 2A](#) presents the molecular structures of the investigated polymers, highlighting their distinct chemical compositions and functional groups that may influence laser absorption, thermal decomposition, and carbonization behavior during the irradiation process ([Table S2](#)).

Digital images of the precursor coatings before and after laser treatment are shown in [Figure 2B](#). Before irradiation, white TiO₂-based coatings containing PVDF, PMMA, and PVA appeared relatively uniform. In contrast, TiO₂ mixed with PE or PDMS showed noticeable agglomeration and poor dispersion, making it difficult to obtain thin and uniform coatings. This behavior can be attributed to the nonpolar and hydrophobic nature of PE and PDMS, which results in poor compatibility with TiO₂ and promotes particle agglomeration.

After exposure to CO₂ laser irradiation, substantial darkening was observed in polymer-containing samples, indicating enhanced photothermal conversion and localized carbonization compared with the no-polymer control. Among the tested materials, the PVDF-containing coating exhibited the most homogeneous and intensely dark surface after irradiation, suggesting more efficient laser-induced transformation and improved energy absorption.

Numerous characterization techniques are used to analyze material properties. Scanning electron microscopy (SEM) was used to investigate the morphology of TiO₂ nanoparticles before and after laser modification. It can be seen that TiO₂ remains spherical before and after laser treatment, with an average size of around 20 nm ([Figure S2](#)). The scanning electron microscopy–energy dispersive X-ray spectroscopy (SEM–EDS) showed the elements present and their distribution. Ti and O are two major elements in white TiO₂ (W-TiO₂) and L- TiO₂ ([Figure S3 and](#)

table S3). The optical properties of the resulting samples were further evaluated by measuring their average solar absorptance over the wavelength range of 300–2500 nm (Figure 2C–Figure S4). The no-polymer (L-TiO₂) samples showed a very low solar absorptance of **18.2%**. In contrast, polymer-assisted samples showed significantly enhanced solar absorption. L-TiO₂/PVDF, L-TiO₂/PVA, and L-TiO₂/PMMA achieved high solar absorption of larger than 90%, outperforming L-TiO₂/PE and L-TiO₂/PDMS. Notably, PVDF combined both high absorptance of 94.3% and superior surface uniformity after laser processing.

For comparison, the white TiO₂/PVDF hybrid with the same PVDF content (10 wt%) was pyrolyzed under Ar at 600 °C for 1 h (denoted as P-TiO₂/PVDF). P-TiO₂/PVDF exhibited a gray color and much lower solar absorption than L-TiO₂/PVDF (94.1%). The enhanced absorption of P-TiO₂/PVDF can be attributed to carbon residues formed during the thermal decomposition of PVDF. In contrast, the significantly higher solar absorption of L-TiO₂/PVDF likely originates from the combined effects of Ti-related defects and carbon species generated during laser treatment. Thermal response measurements under 1 sun illumination are summarized in (Figure 2D–Figure S6). After laser treatment, the surface temperature of L-TiO₂/PVA, L-TiO₂/PMMA, L-TiO₂/PE, L-TiO₂/PDMS, and L-TiO₂/PVDF were remarkable higher than L-TiO₂. Notably, the L-TiO₂/PVDF reached the highest surface temperature, approaching ~68 °C after 10 minutes. The superior photothermal performance of PVDF is attributed to its efficient laser-mediated carbonization and strong broadband solar absorption. These results collectively identify PVDF as the most suitable polymer mediator for CO₂ laser conversion of TiO₂ coatings for solar-thermal applications.

2.3. Materials characterizations

The crystal structure of TiO₂ was explored by X-ray diffraction (XRD) (Figure 3A). In the W-TiO₂ sample, the XRD pattern is composed of the peaks of 25.4°, 36.9°, 37.8°, 48.1°, 53.8°, and 55.2°, corresponding to (011), (013), (020), and (110) lattice plane reflections of the anatase TiO₂ phase.

L-TiO₂ shows a small change in crystal structure from anatase to rutile. When the PVDF is loaded as an additive, three new peaks are located at 27.4, 36.1, and 41.1, corresponding to the (110), (101), and (11) lattice planes in the rutile TiO₂ phase. The XRD results suggest that the rutile TiO₂ phase tends to form from the anatase TiO₂ phase at the high temperatures of the laser process. Raman spectra were obtained to elucidate the structure of the prepared materials (Figure 3B). The peaks are located at the Raman positions of 150, 400, 523, and 646 cm⁻¹, indicating the bonding in the TiO₂ particles. The shift peak from 148.7 cm⁻¹ in white TiO₂ to 157.7 cm⁻¹ in the L-TiO₂/PVDF reflects the changes of Ti-O bonding to oxygen defect. The new peaks at 1374 cm⁻¹ and 1591 cm⁻¹ reflect the formation of graphitic and defect carbon during laser carbonization of the polymer chain.

Figure 3 C-E shows the HRTEM images of W-TiO₂, L-TiO₂, and L-TiO₂/PVDF. All three samples have a crystal structure with a fringe spacing of around 0.35 nm, indicating the TiO₂ anatase (101) lattice plane. The selected-area electron diffraction (SAED) patterns further confirm the crystal structure of TiO₂ (Figure S7). A thin disordered layer around 1- 2 nm can be observed outside the crystal core of L-TiO₂/PVDF. This disordered layer is attributed to lattice defects such as oxygen vacancies and Ti³⁺ [22]. TEM-EDX further confirms the well distribution of C, O, and Ti elements within TiO₂ nanoparticles (Figure S8).

The X-ray photoelectron spectroscopy (XPS) was used to examine the change in surface chemical bonding of TiO₂ induced by the laser treatment process (Figure 3F-H). Overall XPS scanning, Ti and O are the major elements in all three samples. A small amount of Al and F was found in L-TiO₂ and L-TiO₂/PVDF, which resulted from the Al foil substrates (Figure 3G). The XPS spectra of Ti2p show the electronic state of TiO₂. Main peak in W-TiO₂ located at the 458.46 eV position. After laser treatment with PVDF supporting, the slight positive peaks (0.22 eV) indicated that Ti³⁺ was formed under laser irradiation in the presence of PVDF. The XPS O1s in W-TiO₂ and L-TiO₂/PVDF show similar differences trend (Figure 3H). In W-TiO₂, the overall peak is

sharper and can be resolved into 2 peaks at 530.0 and 530.9 eV, corresponding to the Ti–O and Ti–OH bonding. After laser treatment, the Ti–O peaks shift 0.7 eV (from 529.8 to 530.5). These shifts could be a result of the presence of additional chemical states in the samples by virtue of the laser induced oxygen vacancies and the associated Ti^{3+} states, that decrease the Ti–O and Ti–OH bond length [23]. In conclusion, the CO_2 laser treatment was used to form Ti^{3+} and an oxygen defect in white TiO_2 on a PVDF support.

2.4. *Ti defect and oxygen vacancies formation mechanism*

To understand the formation of black TiO_2 by using the CO_2 laser process, the laser irradiation absorption schematic is shown in Figure 4A. Pristine W- TiO_2 exhibits weak absorption of the 10.6 μm CO_2 laser irradiation. A large amount of incident laser is reflected or scattered from W- TiO_2 surface. That limiting the efficient photothermal conversion and generation of high temperature required for defect formation process. In contrast, the introducing PVDF into the TiO_2 layer markedly improves infrared energy absorption. The hybrids W- TiO_2 /PVDF contains several characteristic organic bonds, including C–F, C–C, and C–H groups, which can interact with the laser irradiation through vibrational excitation. Fourier transform infrared spectroscopy (FTIR) analysis reveals the absorption characteristics of PVDF (Figure 4B). From the FTIR spectra, PVDF shows a strong vibrational mode, including bands nears 950 cm^{-1} assigned to C–F stretching/bending, 1050 cm^{-1} assigned to skeletal C–C vibrations, 1175 cm^{-1} are attributed to C–F stretching, and band near 1400 cm^{-1} corresponds primarily to CH_2 bending vibrations. Besides, the CO_2 laser wavelength of 10.6 μm is corresponding to a wavenumber of nearly 943 cm^{-1} , it overlaps with the C–F vibration of PVDF. This overlap enables PVDF to play as an efficient infrared absorber, converting the incident laser energy into thermal energy on the TiO_2 /PVDF hybrid surface. The reaction during laser process illustrated in Figure 4C, the rapid heating induced by laser irradiation promotes PVDF decomposition and generation of fluorine

species and carbonaceous fragments. These reactive species, combined with the highly localized thermal environment, can facilitate oxygen removal from the TiO_2 lattice and consequently generate Ti-related defects and oxygen vacancies. The formation of oxygen-deficient TiO_{2-x} introduces defect states within the electronic structure, which enhances visible and near-infrared light absorption and causes the characteristic color transition from white to black. Therefore, PVDF serves not only as a polymeric additive but also as an infrared sensitizer and chemically active component that promotes localized laser heating and defect formation, enabling the direct conversion of conventional white TiO_2 into black TiO_2 under CO_2 laser irradiation in air.

2.5. Optical properties and solar vapor generation performance

The properties of L- TiO_2 /PVDF show potential for solar vapor generation. In this research, L- TiO_2 /PVDF was mixed PVDF binder 5% to form a homogeneous ink and subsequently coated onto white cotton fabric (L- TiO_2 /PVDF@Ct). The SEM images showed that it has a good distribution on cotton cloth (Figure S9). The hydrophilicity of the materials was evaluated by measuring the water contact angle. In Figure S10, the water droplet was completely absorbed after 0.1 second, resulting in a contact angle of approximately 0° . This rapid water absorption leads to continuous and sufficient water transport from the bulk water to the surface of the photothermal evaporator. Besides, the temperature changes in the evaporation system also monitored using a thermal infrared camera. The evaporation surface temperature for the L- TiO_2 /PVDF@Ct surface rapidly increases within the first 10 minutes and then stabilizes at 35.1°C . To evaluate the performance of the L- TiO_2 /PVDF@Ct system, the mass change of water over time was recorded in darkness and under one-sun irradiation (Figure 5A). Besides, raw cotton (R_ct), laser-induced PVDF-coated cotton (L-PVDF@Ct), and commercial black carbon-coated Ct (BC/Ct) were prepared with the same mass loading as the reference samples. Under darkness, Pure water, R_ct, L-PVDF@Ct, BC@Ct, and L-PVDF@Ct can obtain the evaporation rate of 0.11,

0.17, 0.23 and 0.24 kgm⁻²h⁻¹ (Figure S11). Under 1 sun illumination, the L-TiO₂/PVDF@Ct -based photothermal evaporator can get a high evaporation rate of 1.51 kgm⁻²h⁻¹, which is larger than L-PVDF/Ct, 4 times compared with Raw Ct, and 7 times that of pure water (Figure 5B). Furthermore, L-TiO₂/PVDF@Ct achieves a high energy efficiency of 88% under 1 sun illumination (Figure 5C). Compared with the recently reported flat black-TiO₂-based photothermal evaporator, the L-TiO₂/PVDF@Ct is comparable with almost all previous reports (Figure 5D). The evaporator shows rapid salt rejection, with 1 g NaCl completely removed after 110 minutes (Figure. S12). The desalination performance of L-TiO₂/PVDF@Ct for real seawater was tested (seawater was collected in Geoje Island, Gyeongsang Nam, Korea, on 26 May, 2026). L-TiO₂/PVDF@Ct showed a high desalination performance with an evaporation rate of 1.46 kg.m².h (Figure. S13). After exposure in seawater and outdoor conditions for a long time, the L-TiO₂/PVDF still has good adhesion on cotton. SEM and XRD further confirm the stability of active materials on the cotton substrate (Figure. S14-15). This leads to the long-term stability of desalination performance.

2.6. Outdoor practice application

Outdoor experiments were conducted on the main campus of Korea Institute of Machinery & Materials in Daejeon, Korea. From 09:00 to 17:00 daily, we monitored rooftop solar irradiance and cumulative freshwater yield of L-TiO₂/PVDF@Ct evaporators. On a sunny day, solar flux fluctuated in a range of 0.5 – 0.94 kWm⁻² (Figure 6A). The freshwater collection was 21.04 kg after 7 hours, with the average evaporation rate of 3.01 kgm⁻² h⁻¹ (Figure 6B). To evaluate the condense water quality, we designed a solar desalination device made by transparent PMMA utilizing cooling water to facilitate vapor condensation (Fig. 6D). The L-TiO₂/PVDF@Ct evaporators exhibited remarkable purification ability. The ohmic value of seawater before and after purification were measured with multimeter. The ohmic vale drop from 416 kΩ to 0.76 MΩ indicating the reduction of cation and anion. Furthermore, Key ion concentrations dropped from

9430 (Na^+), 420 (Ca^{2+}), 879 (Mg^{2+}), and 422 (K^+) $\text{mg}\cdot\text{L}^{-1}$ to 1.62, 0.71, 0.20, and 1.15 $\text{mg}\cdot\text{L}^{-1}$, respectively (Figure 6E). reducing feedwater salinity by approximately four orders of magnitude - well below the World Health Organization (WHO) drinking water standards of 1‰ (Figure 6F).

3. Conclusion

Black TiO_2 has emerged as a promising material for solar energy conversion due to its defect-induced visible-light absorption; however, its scalable and rapid synthesis remains challenging. Conventional methods-such as hydrogenation, chemical or electrochemical reduction, and solvothermal processes-often require high temperatures, prolonged reaction times, or the use of additional chemical reagents. Laser-based fabrication offers a clean and rapid alternative; however, the direct CO_2 -laser conversion of white TiO_2 is inefficient due to its strong infrared reflectance. To overcome this limitation, we introduce polyvinylidene fluoride (PVDF) as an infrared-absorbing and chemically reducing support, enabling the rapid transformation of white TiO_2 into black TiO_2 . PVDF strongly absorbs 10.6 μm CO_2 -laser radiation, generating localized photothermal heating while decomposing into reactive species (H , CF_2 , and carbonaceous fragments) that promote Ti^{4+} reduction, oxygen-vacancy formation, and surface disorder. Using a CO_2 laser (60 W, 300 mm s^{-1} scan speed, 0.18 mm line gap), black TiO_2 is successfully synthesized within seconds, exhibiting markedly enhanced visible-light absorption and a high solar absorptance of 94%. Under outdoor sunlight, the material reaches temperatures up to 68 °C. This rapid, solvent-free, and highly efficient approach provides a promising platform for scalable fabrication of defect-engineered TiO_2 for photocatalytic and solar-vapor generation application.

4. Experimental section

4.1. Synthesis of black titanium oxide from white titanium oxide

Preparation of white- TiO_2 /Polymer hybrid coated on Al foil:

To investigate the effect of polymer support, five commonly used white TiO₂/polymer-coated Al foils, including polyvinylidene (PVDF), polyethylene (PE), poly(methyl methacrylate) (PMMA), polydimethylsiloxane (PDMS), and polyvinyl alcohol (PVA), were prepared:

1. The commercial white TiO₂ (W-TiO₂) was mixed with PVDF (9:1 wt%) and N-methyl-2-pyrrolidone (NMP) to form a homogeneous solution, then was cast onto a Al foil (10 μm thickness) with a mass loading of 5mg/cm². The W-TiO₂/PVDF hybrid film was then dried at 100°C overnight to ensure no NMP residue remained.
2. The commercial W-TiO₂ was mixed with PE (9:1 wt%) and xylene (60 °C) to form a homogeneous solution, then was cast onto an Al foil (10 μm thickness) with a mass loading of 5mg/cm². The W-TiO₂/PE hybrid film was then dried at 60°C overnight to ensure no xylene residue remained.
3. The commercial white W-TiO₂ was mixed with PMMA (9:1 wt%) and chloroform to form a homogeneous solution, then was cast onto an Al foil (10 μm thickness) with a mass loading of 5mg/cm². The W-TiO₂/PMMA hybrid film was then dried at 60°C overnight to ensure no chloroform residue remained.
4. The commercial white W-TiO₂ was mixed with PDMS (9:1 wt%) and toluene to form a homogeneous solution, then was cast onto an Al foil (10 μm thickness) with a mass loading of 5mg/cm². The W-TiO₂/PDMS hybrid film was then dried at 60°C overnight to ensure no toluene residue remained.
5. The commercial W-TiO₂ was mixed with PVA (9:1 wt%) and DI water (90°C) to form a homogeneous solution, then was cast onto an Al foil (10 μm thickness) with a mass loading of 5mg/cm². The W-TiO₂/PVDF hybrid film was then dried at 100°C overnight to ensure no water residue remained.

Laser-induced black TiO₂: A CO₂ laser (LP-400 series CO₂ laser maker Panasonic) was used to synthesize black TiO₂. The instrument has the highest power of 75 W and a wavelength of

10.64 μm . The experiment was conducted at room temperature in air in a scan mode at a scan speed of 300 mm/s, with a line gap of 0.18 mm and laser power of 50%. A fume exaction system (TOGA clean system) was used to ensure no toxic production spread out. The samples' names were denoted as L-TiO₂/X, where X represented the polymer supporting.

Pyrolysis W-TiO₂/PVDF hybrids: white TiO₂ was mix with PVDF (10 wt%) in a molar. Subsequently, this hybrid was pyrolysis in furnace under Ar gas (150 sccm), at 600°C for 1 hours. The obtain samples was denoted as P-TiO₂/PVDF

Photothermal membrane preparation for UV-Vis-NIR measurement: L-TiO₂/X (90%), PVDF (10 wt%), and NMP were ground in a mortar to form a homogeneous solution, which was then cast onto a transparent glass substrate and dried at 100 °C overnight. The samples obtained have a mass loading of 4.5 mg/cm².

Fabrication of L-PVDF/X-painted white cotton-based photothermal evaporator.

L-TiO₂/X and PVDF 5 wt% were ground in a mortar to form a homogeneous solution. The obtained solution was coated on the white cotton and dried at 100°C overnight. The final photothermal evaporator has a mass loading of around 4.5 mg/cm².

4.2. Evaluation of solar-driven vapor evaporation of the L-TiO₂/X@ct

The solar-driven evaporation performance of the L-TiO₂/X@ct composite was systematically evaluated using a solar simulator (K201 LAB 55 with K401 CW150 Lamp Controller, McScience) calibrated to deliver a simulated solar flux of 1 kW m⁻². Before testing, the irradiance was verified using a solar power meter (TES-132, TES), and the beam aperture was adjusted to fully illuminate the sample surface with a 5% overshoot to account for edge effects. The evaporator assembly consisted of L-TiO₂/X@ct mounted on hydrophilic cellulose tissue affixed to a 2 cm-thick polystyrene foam base. This configuration served dual purposes: (1) minimizing conductive heat loss to the bulk water by thermally decoupling the evaporator, and (2) enabling continuous

capillary-driven water transport through the porous cellulose matrix. Real-time thermal profiling was conducted using a high-sensitivity infrared camera (A325sc, FLIR) for surface temperature mapping, complemented by bulk water temperature monitoring via a T-type thermocouple (Center309, CENTER).

Mass loss measurements were performed at one-second intervals using a precision analytical balance (ADB 200-4, KERN; resolution: 0.1 mg). All experiments were conducted under tightly controlled environmental conditions ($23 \pm 1^\circ\text{C}$, $57 \pm 3\%$ RH) to eliminate ambient variability. The solar-to-thermal conversion efficiency (η , %) was determined using the following formula:

$$\eta = \Delta m (h + Q) / P$$

Where Δm is the evaporation rate with the dark evaporation rate subtracted ($\text{kg m}^{-2} \text{h}^{-1}$), h is the evaporation enthalpy of water (kJ kg^{-1}), Q is the sensible heat of water, and P is the solar illumination intensity ($3600 \text{ kJ m}^{-2} \text{h}^{-1}$).

4.3. Water desalination and wastewater treatment

To assess the desalination capabilities of the L-TiO₂/PVDF@ct system, simulated saltwater with a concentration of 3.5 wt% was prepared by dissolving 35 g of NaCl in 965 g of deionized water. For practical application, real seawater was collected in Geoje island, Gyeongsang, Korea. The salinity of all samples before and after desalination was monitored by a salinity refractometer (0 to 100 ppt) with ATC. The concentration of cations in the solutions was measured both before and after the desalination and purification processes using Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES, Ethos 1, Milestone) to ensure precise quantification.

4.4. Evaluation antibacterial activity of L-TiO₂/PVDF

E. coli bacterial was cultured onto LB agar and incubating plated overnight at 37 °C. Bacteria (E. coli MG1655, B. subtilis PY79, and P. aeruginosa PAO1) were cultured by initially streaking glycerol stocks stored at –80 °C onto LB agar and incubating plates overnight at 37 °C

4.5. Materials characterization

The morphology of the samples was characterized by a scanning electron microscope (SEM, JSM-7800F, JEOL). The elemental composition was studied by energy-dispersive X-ray spectroscopy (EDX, AZtec® X-Max, Oxford Instruments). The phase structure and chemical bonds were analyzed by X-ray powder diffraction (Cu K α , Empyrean, PANalytical), and X-ray photoelectron spectroscopy (NEXSA G2, Thermo Fisher Scientific). UV-Vis-NIR spectrometer (LAMBDA 1050, Perkin Elmer) equipped with an integrating sphere was used to measure the light absorption of the solar absorbers. The microstructure of the samples was studied by transmission electron microscopy (Tecnai F30, FEI, 300 kV). The elemental analyses of the individual nanoparticles were performed by STEM-EDX (Talos F200X, FEI, 200 kV). FTIR spectroscopy used the Bruker model (VERTEX 80V) with a scanning range from 400 to 4000 cm⁻¹.

Conflict of interest

The authors declare no conflict of interest.

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FIGURES

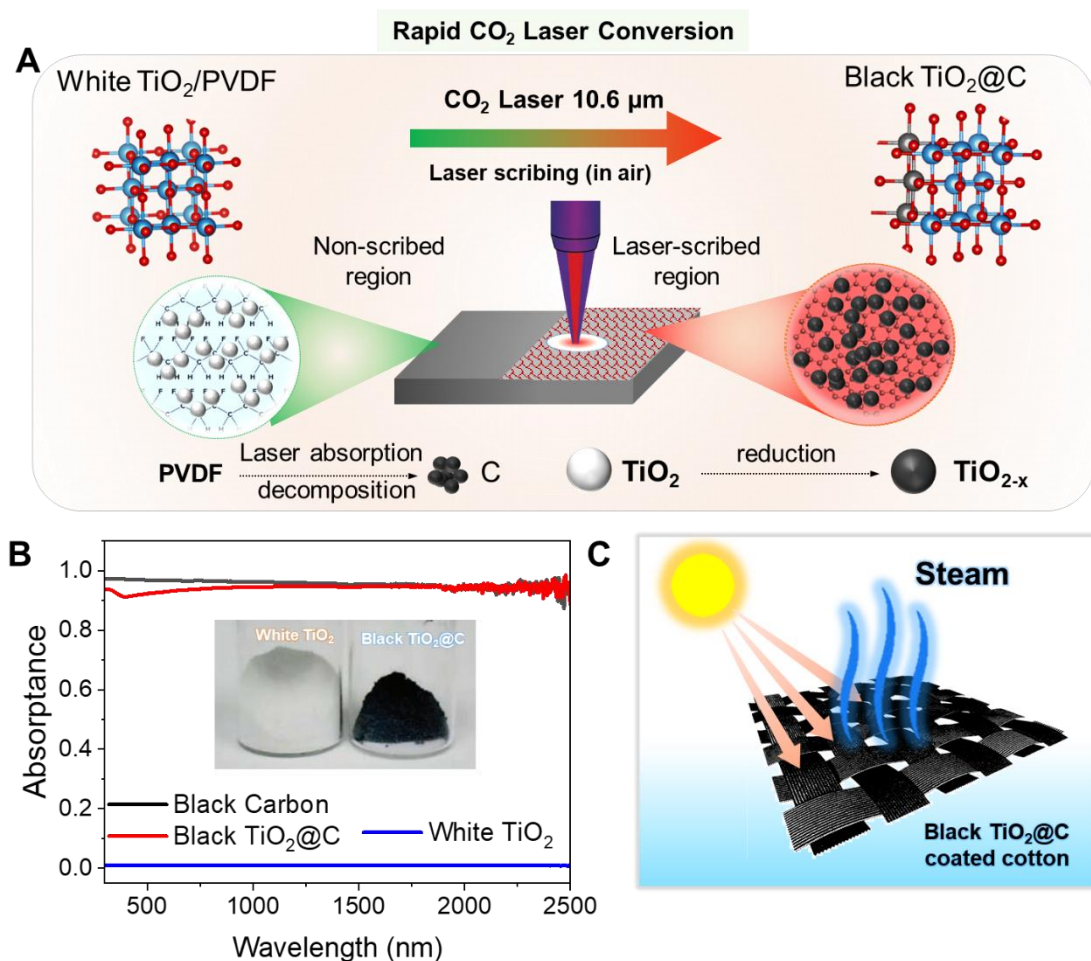


Figure 1. Materials preparation. (A) Schematic illustration of rapid CO₂ laser irradiation (10.6 μm) of TiO₂/PVDF coated on Al foil in ambient air, leading to the formation of hybrid black TiO₂@C through localized photothermal heating, PVDF-assisted reduction, and carbonization. **(B)** Broadband optical absorptivity of the laser-converted sample (L-TiO₂/PVDF) compared with pristine white TiO₂ (W-TiO₂) and commercial black carbon, demonstrating enhanced light harvesting across the UV–Vis–NIR region. **(C)** Schematic of the black TiO₂@C-coated cotton evaporator for efficient solar-driven vapor generation and freshwater production.

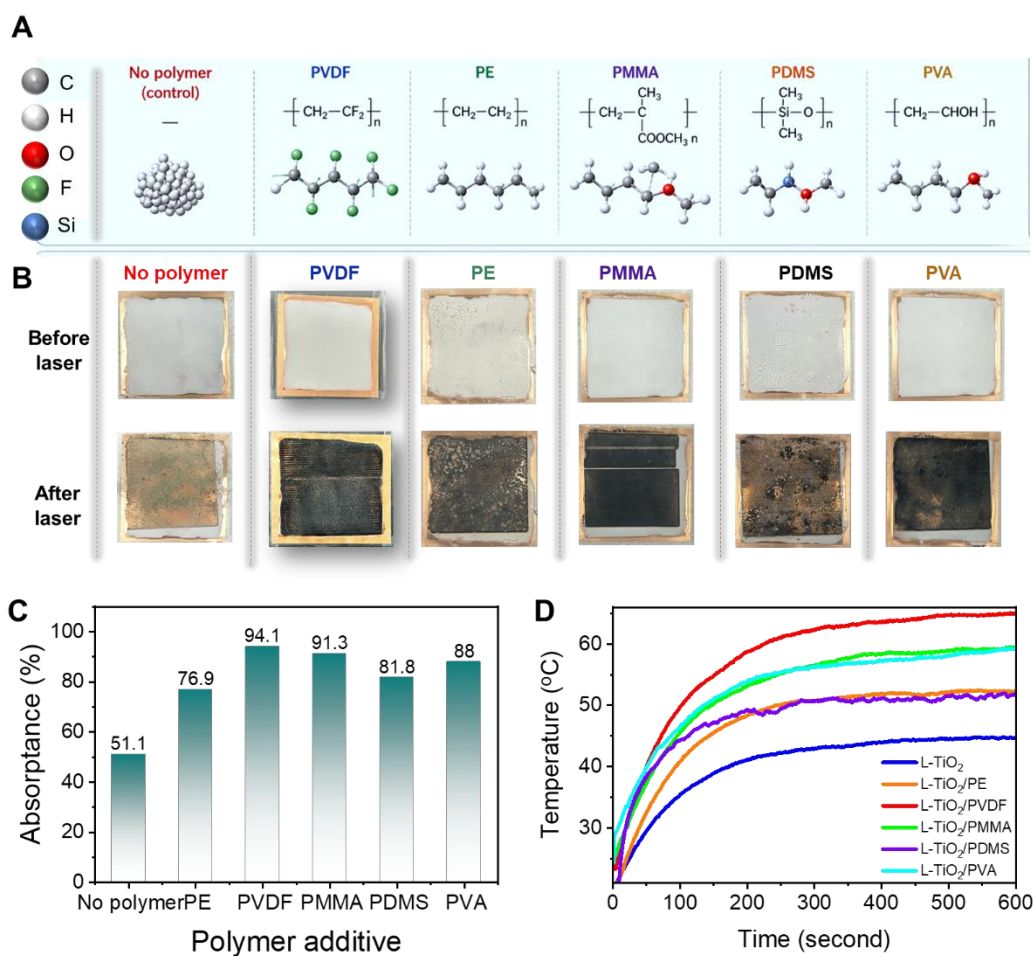


Figure 2. Polymer screening identifies PVDF as the optimal mediator for CO₂ laser conversion of TiO₂. **(A)** The chemical structures of representative polymers and the no-polymer control used for laser treatment under identical conditions. **(B)** Optical photographs of TiO₂/polymer precursor coatings before and after CO₂ laser irradiation (10.6 μ m) in ambient air. **(C)** Average solar absorptance (300–2500 nm) of the resulting samples. **(D)** Temperature profile of surface samples under 1 sun irradiation.

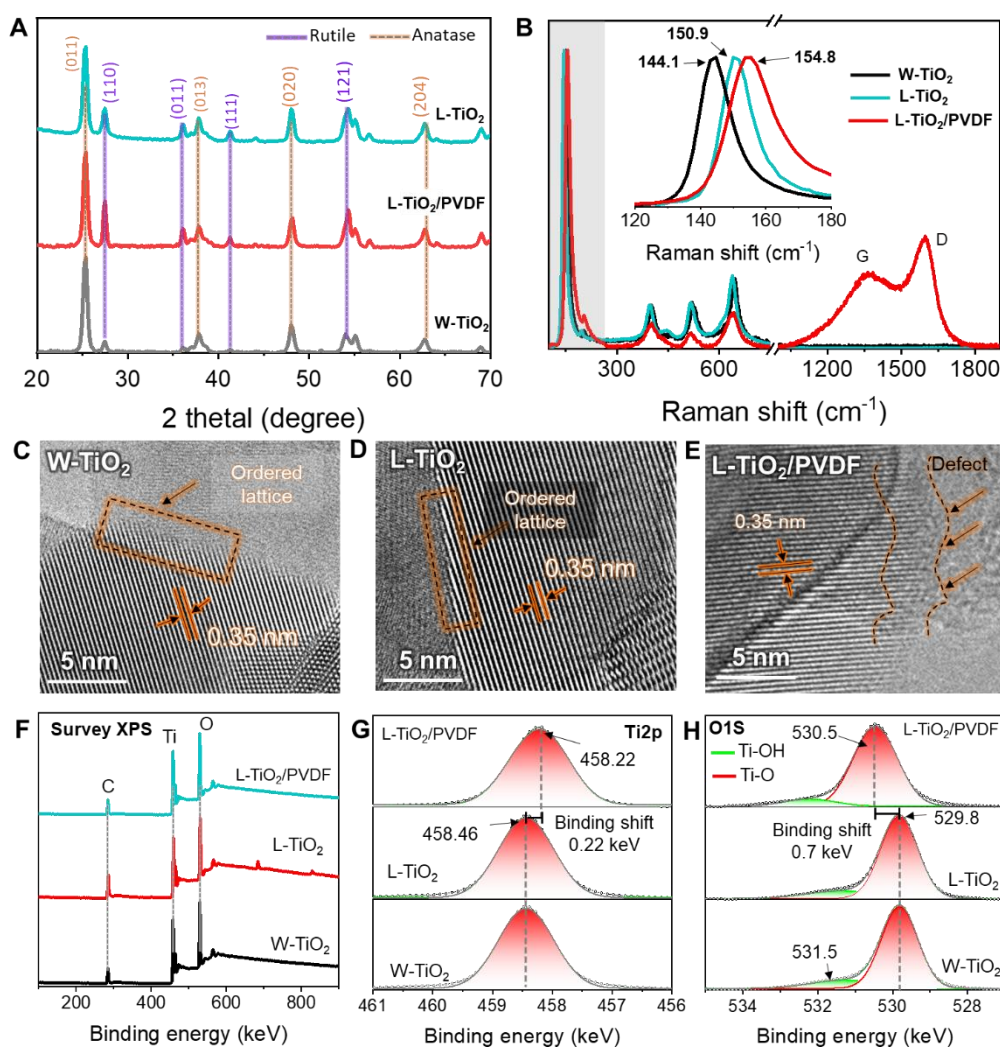


Figure 3. Materials characterization of laser-processed TiO_2 samples. (A) XRD patterns confirming that the anatase/rutile phases are retained after laser processing. **(B)** Raman spectra showing a slight shift of the Eg mode, indicating lattice distortion induced by laser irradiation. **(C–E)** HRTEM images showing ordered lattice in pristine samples and defect formation in L- TiO_2 /PVDF. **(F)** XPS survey spectra confirming elemental composition. **(G)** High-resolution Ti 2p spectra. **(H)** O 1s spectra.

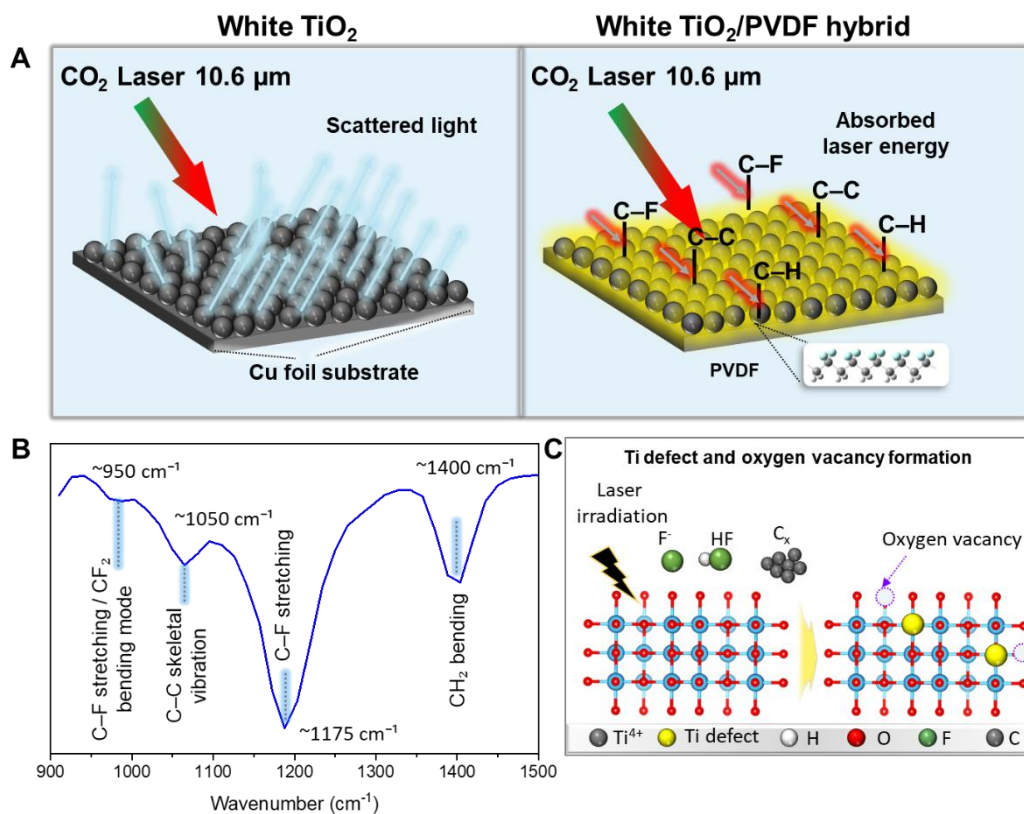


Figure 4. Mechanism of PVDF-assisted infrared absorption under CO₂ laser irradiation. (A) Schematic illustration comparing the interaction of CO₂ laser irradiation on pristine TiO₂ and TiO₂/PVDF hybrid. **(B)** The FTIR spectrum of PVDF highlighting characteristic absorption bands associated with C–F, C–C bonding, which contribute to CO₂ laser irradiation absorption. **(C)** Proposed formation mechanism of Ti defects and oxygen vacancies during laser treatment of TiO₂ /PVDF.

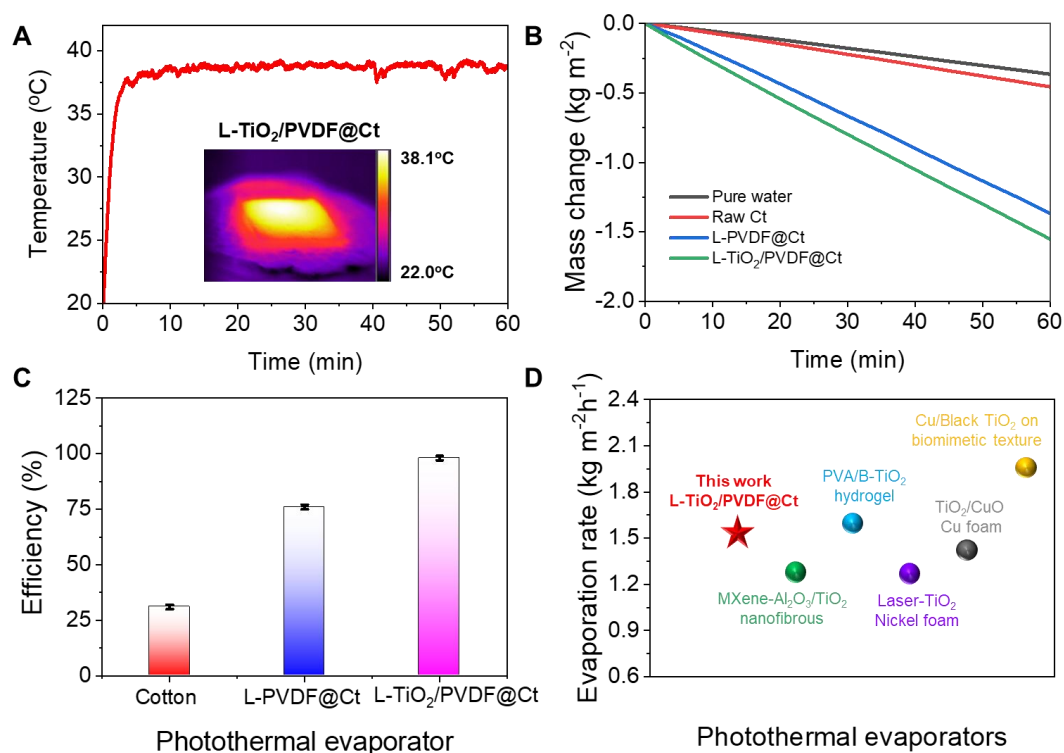


Figure 5. Solar-driven evaporation performance of the optimized L-TiO₂/PVDF@Ct photothermal evaporator. (A) Surface-temperature evolution of L-TiO₂/PVDF@Ct under 1 sun irradiation, with the inset showing the corresponding infrared thermal image. **(B)** Time-dependent mass change of pure water and different evaporator configurations under solar irradiation, demonstrating the enhanced water evaporation achieved by L-TiO₂/PVDF@Ct. **(C)** Calculated solar-to-vapor conversion efficiencies of cotton, L-PVDF/Ct, and L-TiO₂/PVDF@Ct, showing the highest efficiency for the TiO₂/PVDF-based evaporator. **(D)** Comparison of the evaporation rate of L-TiO₂/PVDF@Ct with representative TiO₂-based photothermal evaporators reported in the literature.

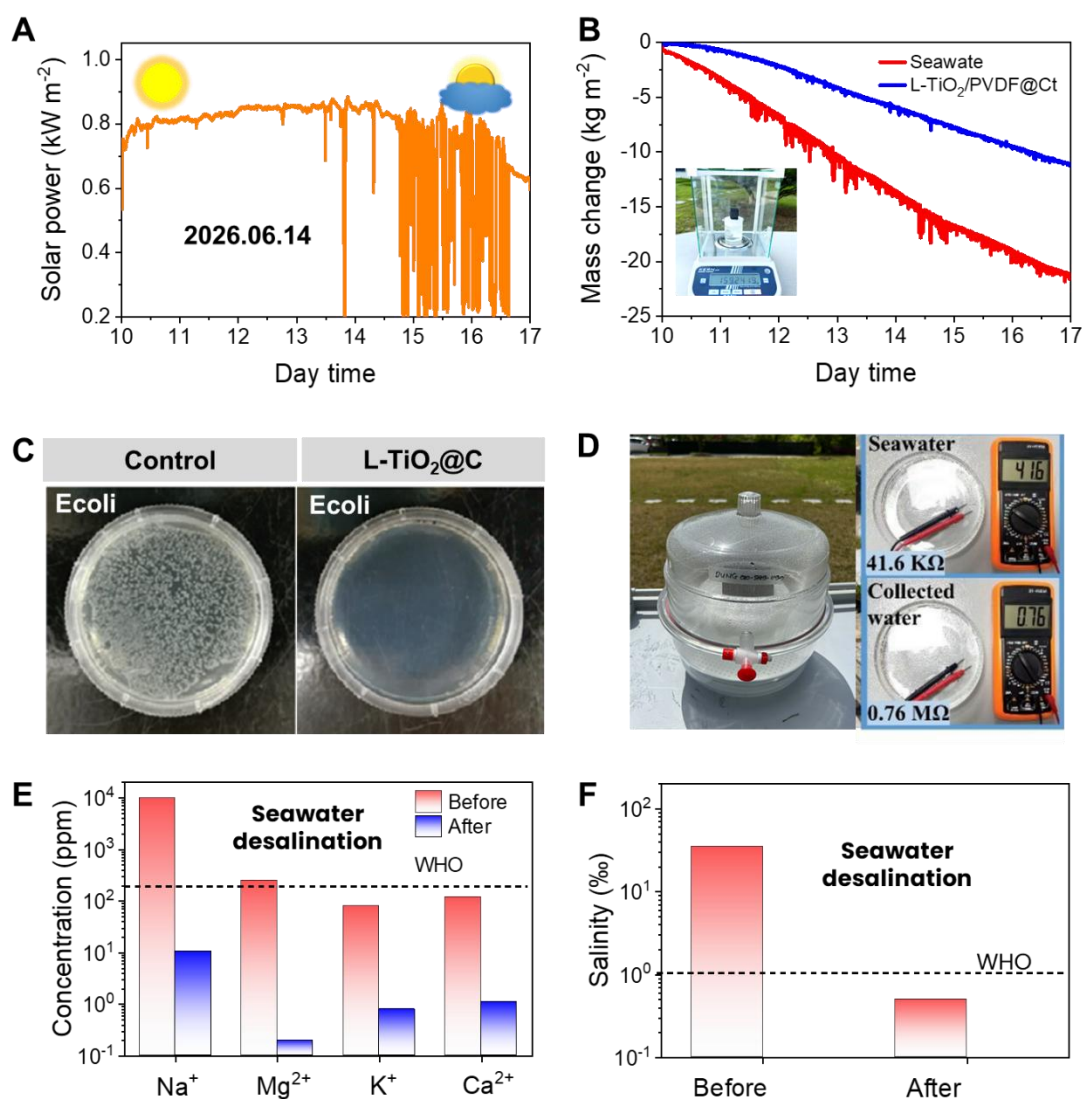


Figure 6. Outdoor desalination performance. (A) Variation in solar power density during the outdoor desalination experiment on June 14, 2026. (B) Cumulative mass change of seawater with and without the L-TiO₂@C evaporator under outdoor solar irradiation. (C) Photographs of *E. coli* colonies for the control and L-TiO₂@C-treated samples, demonstrating the antibacterial performance of the evaporator. (D) Outdoor water-collection setup and electrical resistance measurements of the initial seawater and collected water, indicating a reduction in ionic content after desalination. (E) Concentrations of major ions (Na⁺, Mg²⁺, K⁺, and Ca²⁺) before and after desalination, compared with the WHO guideline. (F) Salinity of seawater before and after desalination, confirming effective salt removal and production of low-salinity collected water.

Supporting information

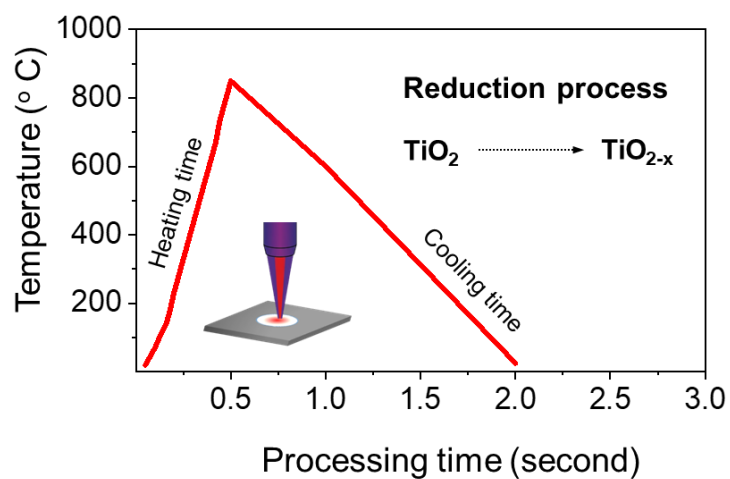


Figure S1. The temperature profile during CO₂ laser processing of the TiO₂/PVDF thin film. The short-lived high-temperature environment promotes the reduction of TiO₂ and the formation of oxygen-deficient TiO_{2-x} species.

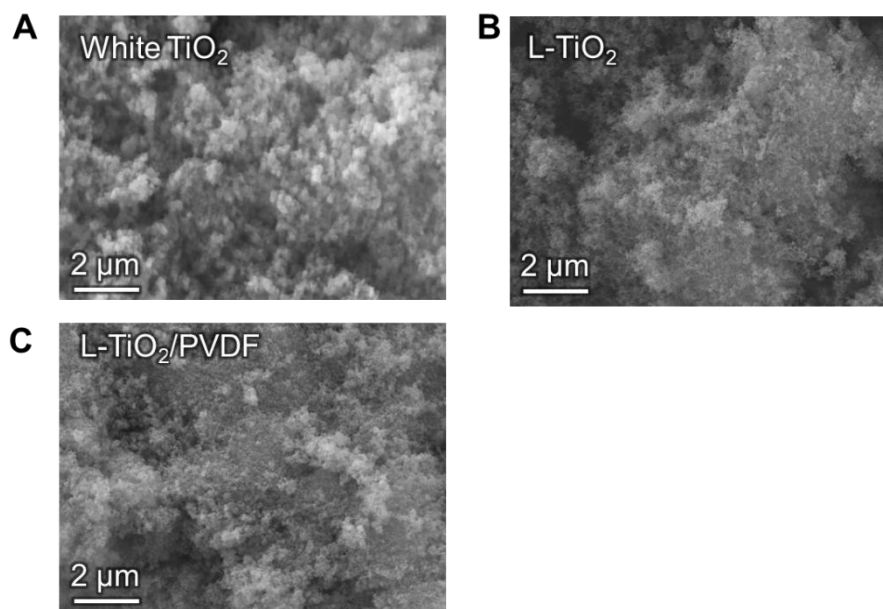


Figure S2. SEM images show the morphology of the prepared samples before and after the laser treatment process.

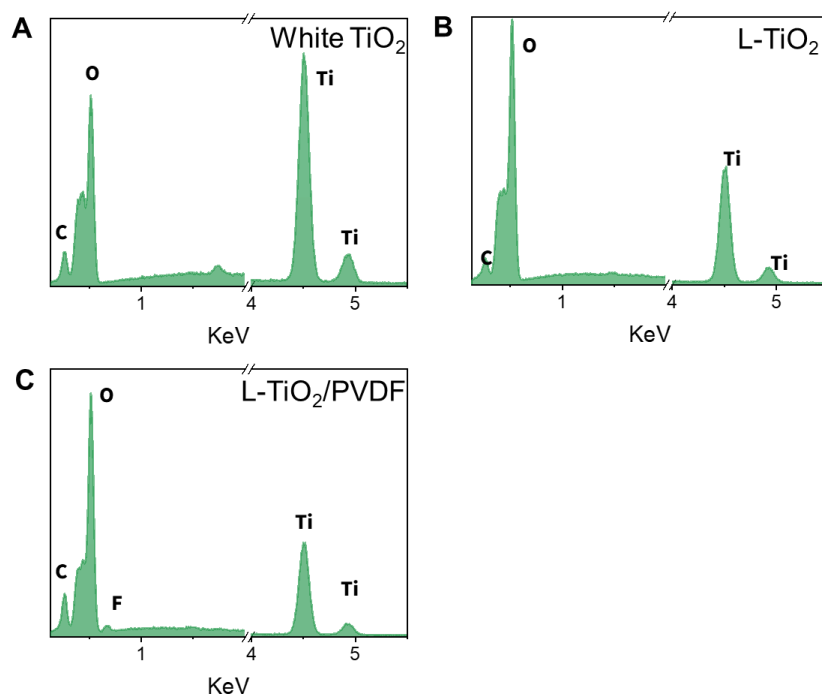


Figure S3. EDS spectra of: (A) pristine white TiO_2 , (B) laser-treated TiO_2 (L- TiO_2), and laser-treated TiO_2 prepared with PVDF as additive.

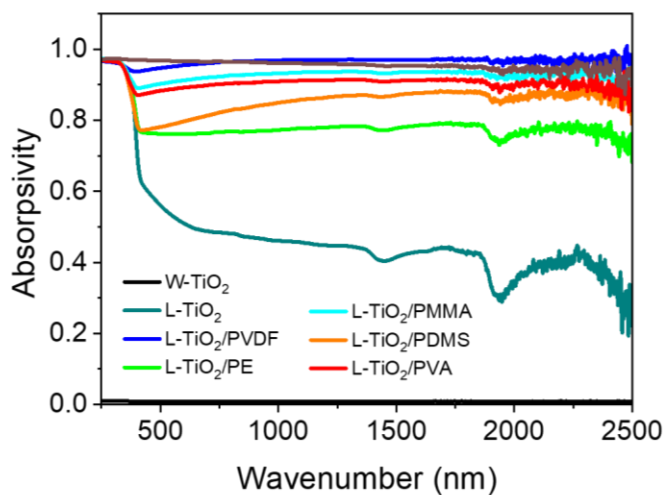


Figure S4. UV-Vis-NIR absorptivity spectra of: pristine white TiO_2 (W- TiO_2), laser-treated TiO_2 (L- TiO_2), and laser-treated TiO_2 prepared with different polymer additives, including PVDF, PE, PMMA, PDMS, and PVA, over the wavelength range of approximately 300–2500 nm.

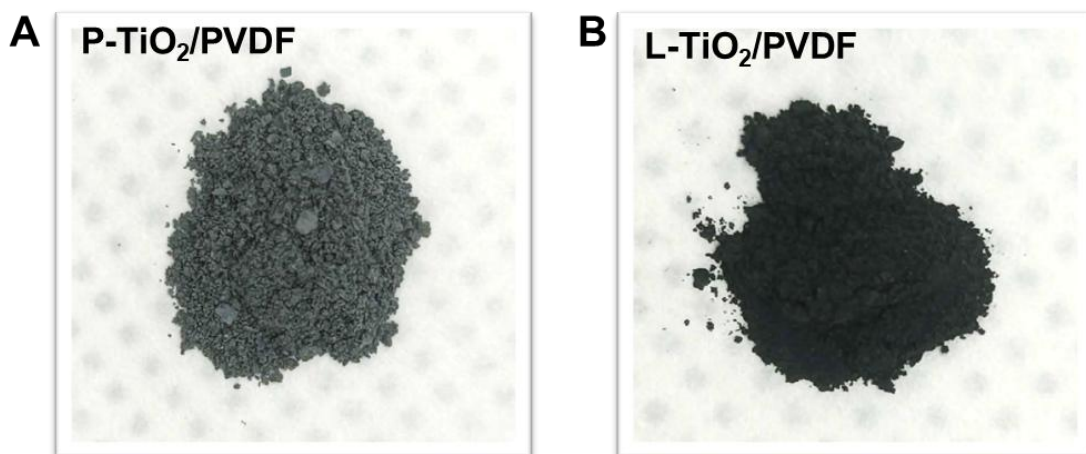


Figure S5. Photographic images of white TiO₂/PVDF hybrid after treatment: (A) pyrolysis at 600°C under Ar gas flow. (B) laser in air under a CO₂ laser irradiation.

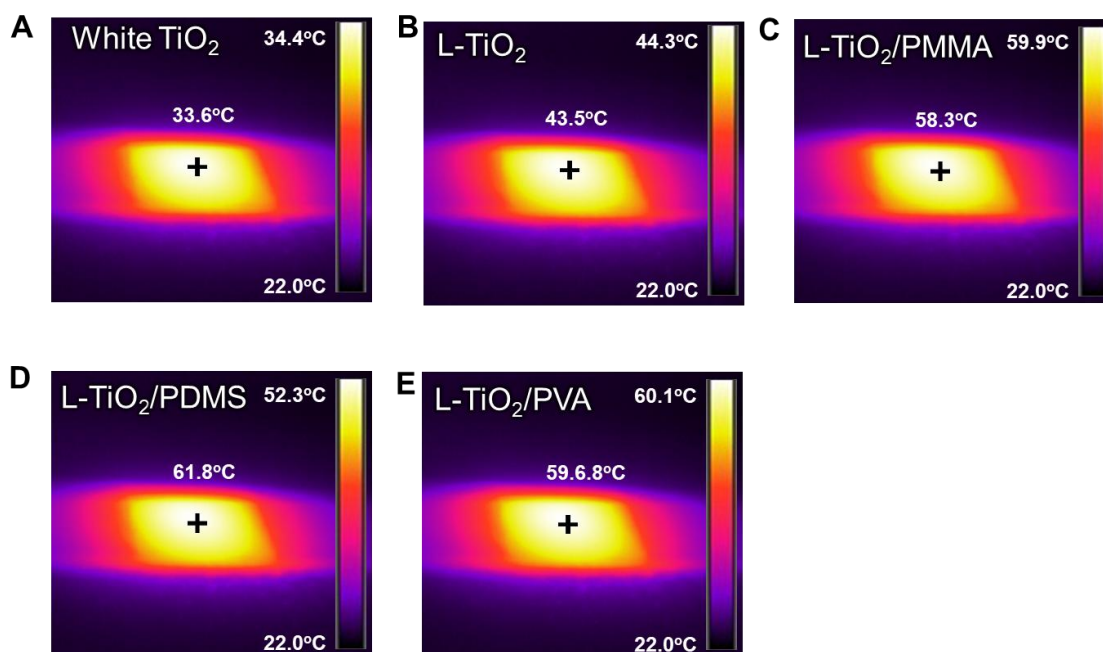


Figure S6. Infrared thermal images of pristine white TiO₂(W-TiO₂), laser-treated TiO (L-TiO₂), and L-TiO₂ prepared with different polymer additives (PVDF, PE, PMMA, PDMS, and PVA) under identical irradiation conditions.

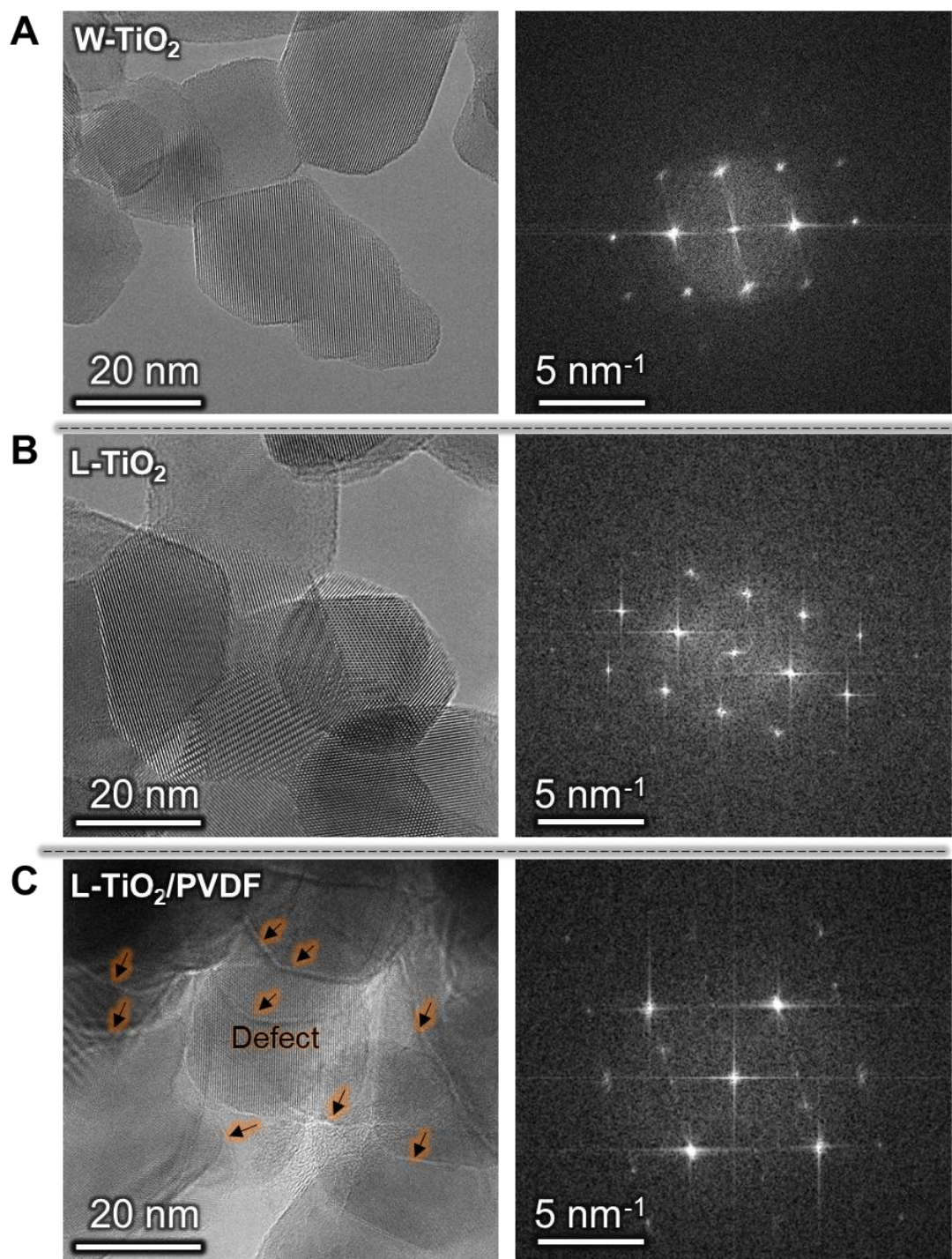


Figure S7. TEM images and corresponding fast Fourier transform (FFT) patterns of (A) pristine white TiO₂ (W-TiO₂), (B) laser-treated TiO₂ (L-TiO₂), and (C) laser-treated TiO₂ with PVDF (L-TiO₂/PVDF)

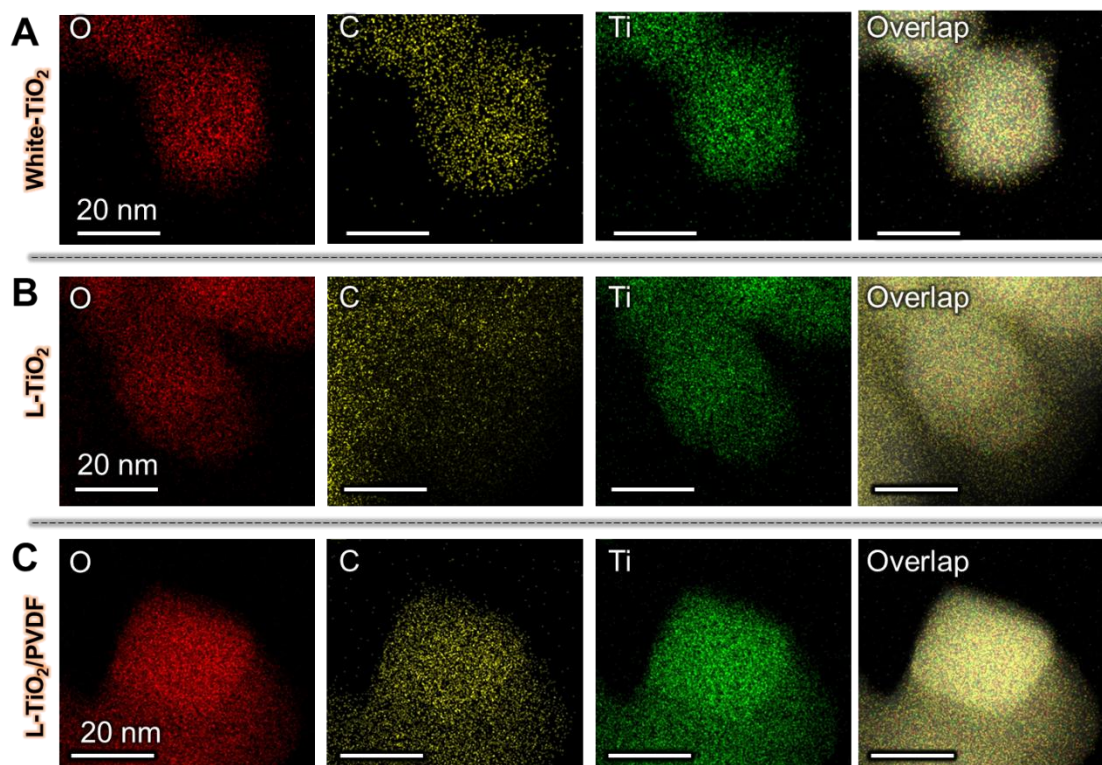


Figure S8. TEM-EDS shows the elemental distribution of: (A) WhiteTiO₂. (B) L-TiO₂. (C) L-TiO₂/PVDF.

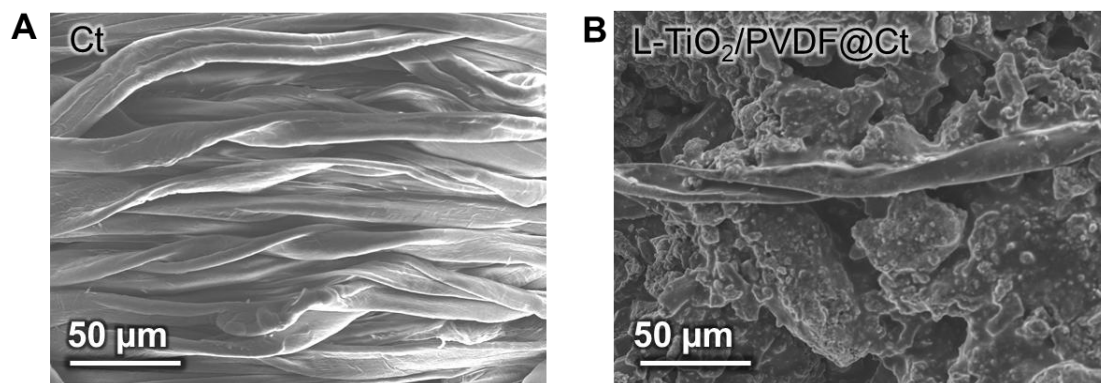


Figure S9. SEM images show the morphology changes before and after coating: (A) Raw white cotton (Ct). (B) L-TiO₂/PVDF coted on cotton (TiO₂/PVDF@ct).

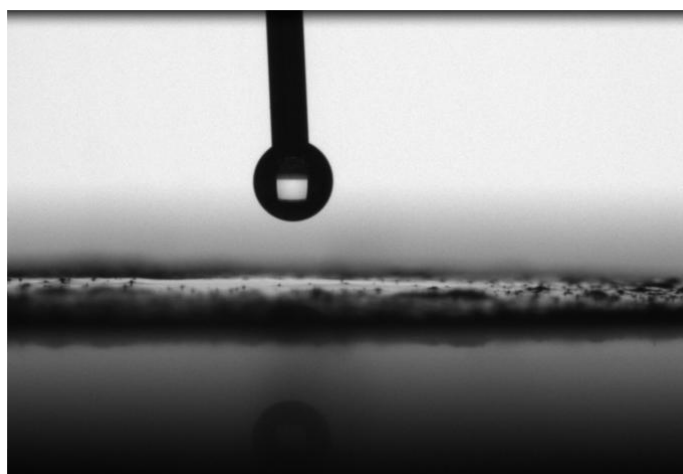


Figure S10. Water contact angle on L-TiO₂/PVDF@ct.

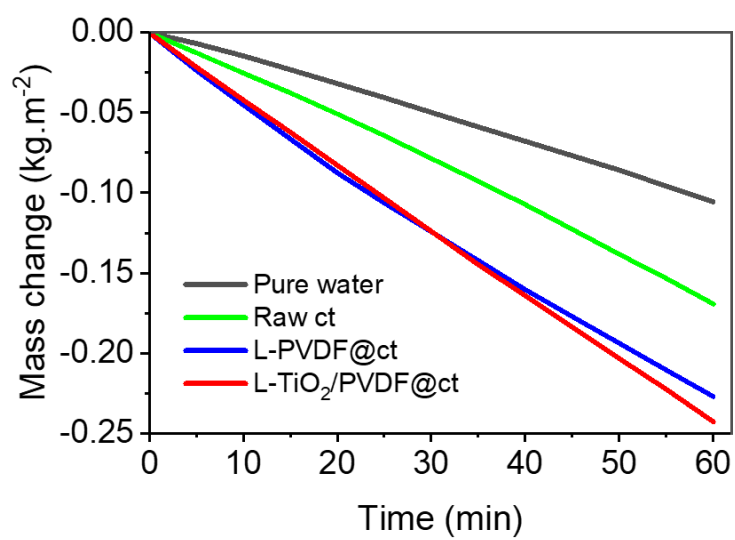


Figure S11. Mass loss of water over time under the darkness of samples.

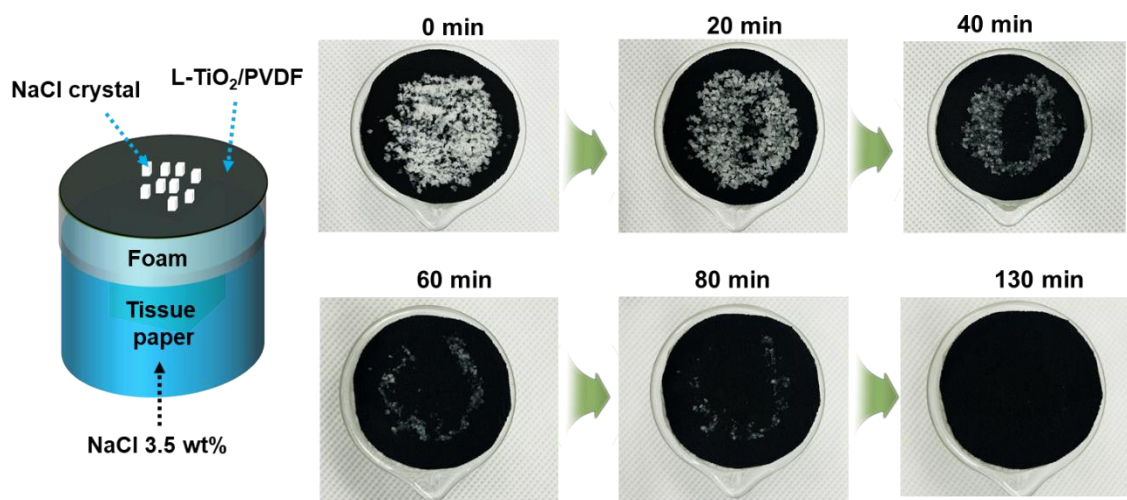


Figure S12. Salt rejection capability showing complete dissolution of NaCl crystals on a wet L-TiO₂/PVDF@ct surface within 130 minutes under one-sun irradiation

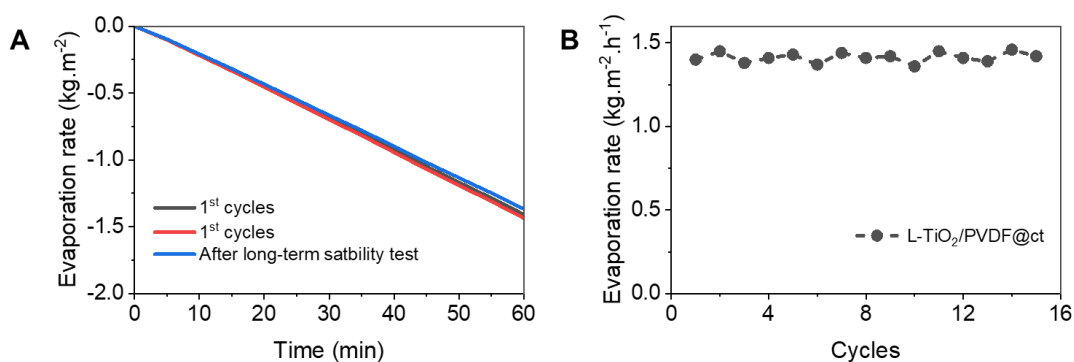


Figure S13. Desalination performance of L-TiO₂/PVDF@ct under one-sun illumination **(A)** The mass change of seawater of L-TiO₂/PVDF@ct as function of time, **(B)** Long-term stability of L-TiO₂/PVDF@ct

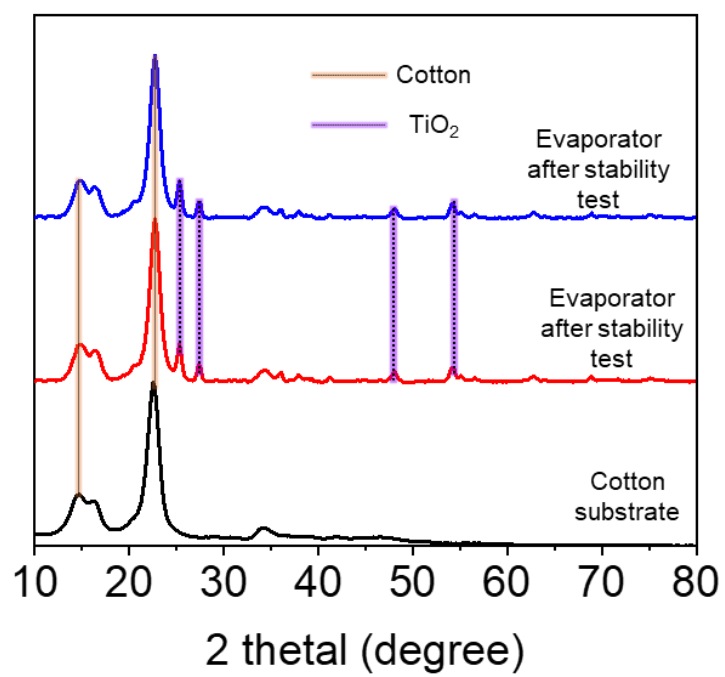


Figure S14. XRD patten of photothermal before and after stability test in real seawater.

Table S1: The advantages and limitation of black-TiO₂ synthesis methods.

Synthesis methods	Advantages	Disadvantages	Ref
TiO ₂ + PVDF	Fast synthesis time Non-solvent requirement	High equipment cost Mostly for thin films Limited scalability	This work
Laser in a wet environment	Highly localized and precise Rapid time Good for patterned surfaces	High equipment cost Mostly for thin films Limited scalability	[18–20]
Hydrogenation (H ₂ reduction)	Produces strong surface disorder Enhances visible/NIR light absorption Significant photocatalytic improvement	High temperature and H ₂ → safety risks Hard to control defect depth Possible particle sintering	[24,25]
Chemical reduction (NaBH ₄ , ...)	Milder conditions than hydrogenation Tunable Ti ³⁺ and oxygen vacancy concentration Works at relatively low temperatures	Air-sensitive reductants Impurity contamination Requires extensive washing	[15]
Plasma treatment	Fast, minutes-scale High defect concentration Low temperature avoids sintering	Requires specialized plasma equipment Limited scalability Potentially nonuniform treatment	[26–28]
Electrochemical reduction	Room-temperature process Precise control of reduction level Environmentally friendly	Mainly suited for films or small batches Shallow reduction depth Needs a controlled electrochemical setup	[29]
Solvothermal/ Hydrothermal Reduction	Easy and scalable Relatively low temperature Allows morphology + defect tuning	Long reaction time Potential carbon contamination Lower reduction level vs strong reductants	[30–32]

Table S2: The properties of polymer screening

Polymer	Main Functional Group / Bonding	Surface Polarity	Solvent
PVDF	Fluoropolymer (–CH ₂ –CF ₂ –), strong C–F dipole	High	NMP
PE	Alkyl chains (–CH ₂ –), van der Waals interaction	Non-polar	Xylene (hot)
PMMA	Ester group (–COOCH ₃), dipole interaction	Moderate	Chloroform
PDMS	Siloxane backbone (Si–O–Si) with methyl groups	Very low	Toluene
PVA	Hydroxyl groups (–OH), hydrogen bonding	High	Hot water

Table S3: Elemental composition by EDS.

Samples	Element					
	C [wt%]	O [wt%]	F [wt%]	Si [wt%]	Ti [wt%]	Al [wt%]
W-TiO₂	0.75 ± 0.11	46.93 ± 0.41	0 ± 0	0 ± 0	52.31 ± 0.51	0 ± 0
L-TiO₂	0.82 ± 0.32	46.72 ± 1.69	0 ± 0	0 ± 0	52.45 ± 2.01	0 ± 0
L-PVDF	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0
L-TiO₂/PVDF	6.08 ± 0.65	46.41 ± 2.97	1.87 ± 0.32	0 ± 0	46.69 ± 0	0 ± 0
L-TiO₂/PE	7.23 ± 2.14	50.48 ± 3.92	0 ± 0	0 ± 0	42.06 ± 3.80	0.22 ± 0.15
L-TiO₂/PMMA	2.69 ± 0.14	45.17 ± 1.09	0 ± 0	0 ± 0	51.74 ± 0.98	0.41 ± 0.03
L-TiO₂/PDMS	3.84 ± 1.13	43.52 ± 9.81	0 ± 0	1.13 ± 1.23	51.51 ± 9.51	0 ± 0
L-TiO₂/PVA	6.87 ± 0.59	48.74 ± 6.09	0 ± 0	0 ± 0	44.15 ± 5.75	0.24 ± 0.07

Table S4: Comparison of SVG performances of various evaporators in the literature.

Evaporator	Evaporation rate	Solar-thermal conversion efficiency	Ref.
	[Kg m ⁻² h ⁻¹]	[%]	
Laser induced black TiO₂ coated cotton (L-TiO₂/PVDF@ct)	1.51 ± 0.05	87%	This work
Cu/Black TiO ₂ on biomimetic texture	1.94	98.56	[33]
Black TiO ₂ on carbon cloth	2.12	66.29	[34]
Superior photothermal black TiO ₂ with random size	1.16	77.14	[35]
MXene-PVA-TiO ₂ -based photothermal	1.23	85.00	[36]
MXene-decorated flexible Al ₂ O ₃ /TiO ₂ nanofibrous	1.43	102	[37]

Laser TiO ₂ on Nickel foam	1.25	78.5	[38]
TiO ₂ -PANidecorated bilayer melamine foam (TiO ₂ -PANi@MF)	2.12	88.9	[39]
MXene-TiO ₂ -Ag with biomass hydrogel	3.09	93.16	[40]
TiO ₂ Films Grown on a Glassy Carbon Foam	2.23	67.1	[41]
PVA/B-TiO ₂ hydrogel	1.60	77.76	[42]
Sawdust hydrochar/titanium dioxide composite (SHC/TiO ₂)	1.29	76.3	[43]
TiO ₂ -CuO-Cufoam	1.28	86.6	[44]
TiO ₂ /NH ₂ -MIL-125(Ti) nanorod array on carbon cloth	1.86	89.5	[45]

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